

FUNDAMENTALS OF DIGITAL ELECTRONICS

EE 8306

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Lecture 01

INTRODUCTION TO DIGITAL ELECTRONICS

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Quiz

- Use Time diagram defined Analogue signal. (2 Marks).
- List four 4 advantage of analogue signal. (4 Marks).
- List four 4 disadvantage of analogue signal. (4 Marks).



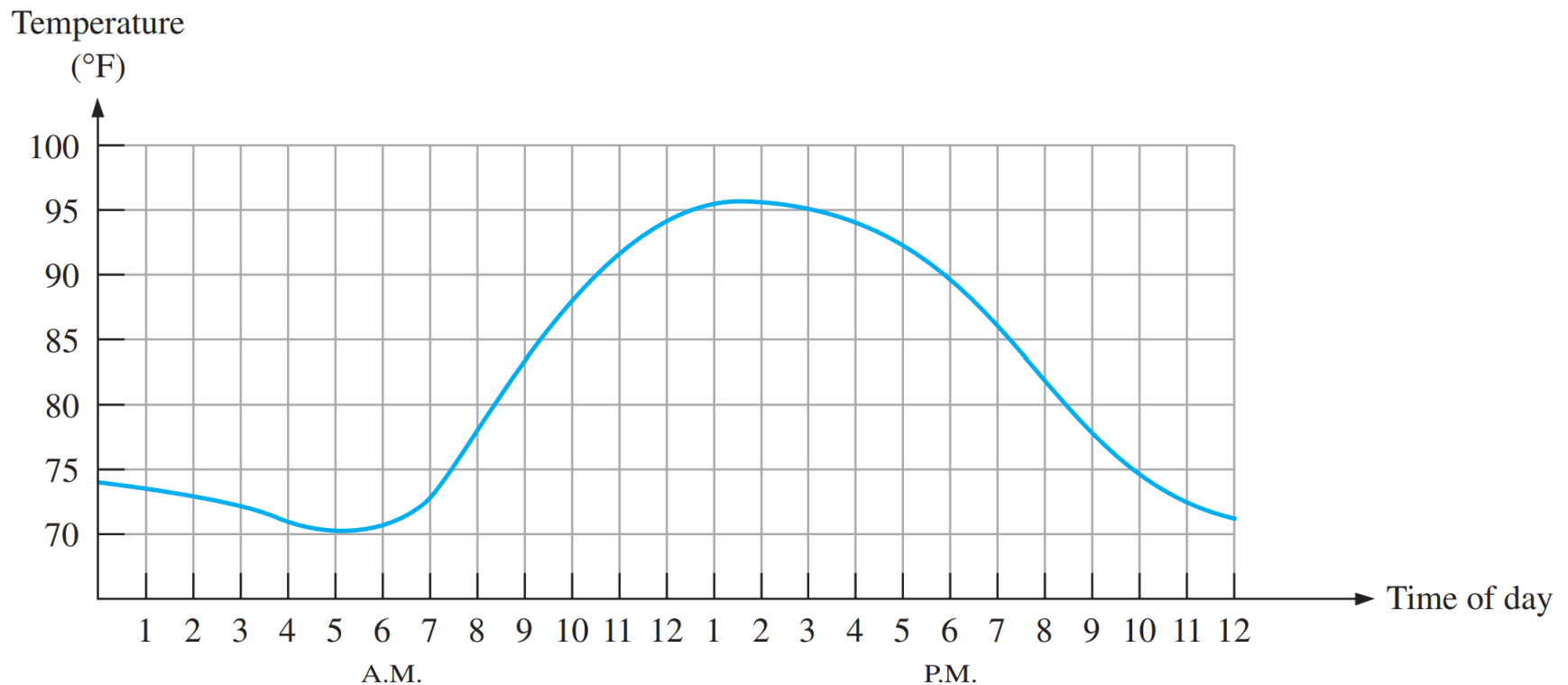
Introduction

- Electronic circuits can be divided into two broad categories,
 - **Digital**
 - **Analog**
- Digital electronics involves **quantities with discrete values**, and analogy electronics involves **quantities with continuous values**.
- An **analog** quantity is one having continuous values.
- A **digital** quantity is one having a discrete set of values.
- Most things that can be measured quantitatively occur in nature in analog form. For example, the air temperature changes over a continuous range of values. During a given day, the temperature does not go from, say, 70° to 71° instantaneously; it takes on all the infinite values in between.



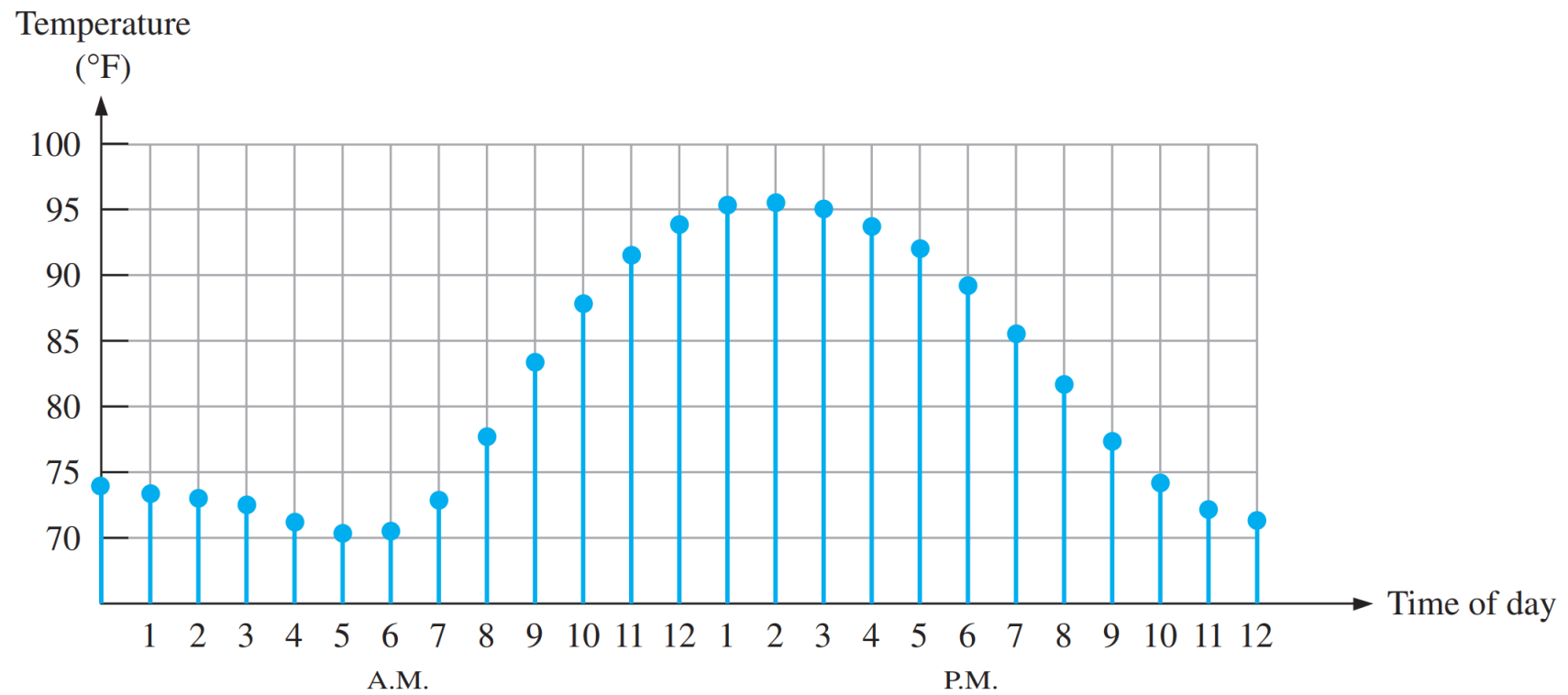
Introduction

- If you graphed the temperature on a typical summer day, you would have a smooth, continuous curve



Introduction

- Rather than graphing the temperature on a continuous basis, suppose you just take a temperature reading every hour.



Introduction

- Now you have sampled values representing the temperature at discrete points in time (every hour) over a 24-hour period. Sampled-value representation (quantization) of the analog quantity in Figure above.
- Each value represented by a dot can be digitized by representing it as a digital code that consists of a series of 1s and 0s.



Advantage of Digital

- Digital representation has certain advantages over analog representation in electronics applications.
 1. Digital data can be **processed** and **transmitted more efficiently** and **reliably** than analog data.
 2. Digital data has a great advantage when **storage** is necessary. For example, music when converted to digital form can be stored more compactly and reproduced with **greater accuracy** and **clarity** than is possible when it is in analog form.
 3. Compared to analog signals, digital data **is not nearly as affected by noise** (unwanted Signal).



Disadvantage of Digital

- Sampling may cause **loss** of information.
- It demands **A/D** and **D/A**, mixed-signal hardware.
- Processor **speed** is limited.
- Develop **quantization** and round-off errors.
- Systems and processing is **more complex**.
- A **higher bandwidth** is required for data communication in comparison to analog transmission of an equivalent information.
- Digital systems and processing are typically **more complex**.



Introduction

- Digital systems are used extensively in **computation** and **data processing, control systems, communications**, and **measurement**.
- Digital systems are capable of **greater accuracy** and **reliability** than analog systems, many tasks formerly done by analog systems are now being performed digitally.
- In a digital system, the physical quantities or signals can assume **only discrete values**, while in analog systems the physical quantities or signals may **vary continuously over a specified range**.
- For example, the output voltage of a digital system might be constrained to take on only two values such as **0** volts and **5** volts, while the output voltage from an analog system might be allowed to assume any value in the range **-10** volts to **+10** volts.



Introduction

- Digital systems work with discrete quantities, in many cases they can be designed so that for a given input, the **output is exactly correct**. For example, if we multiply two 5-digit numbers using a digital multiplier, the 10-digit product will be correct in all 10 digits.
- On the other hand, the output of an analog multiplier might have **an error ranging** from a fraction of one percent to a few percent depending on the accuracy of the components used in construction of the multiplier.
- We can redesign the digital multiplier to process more digits and add more digits to its input.
- A similar improvement in the accuracy of an analog multiplier would not be possible because of limitations on the accuracy of the components.



Introduction

- The **design of digital systems** may be divided roughly into three parts
 - ✓ **System design.**
 - ✓ **Logic design.**
 - ✓ **Circuit design.**
- **System design** involves **breaking the overall system into subsystems** and **specifying the characteristics of each subsystem**. For example, the system design of a digital computer could involve specifying the number and type of memory units, arithmetic units, and input-output devices as well as the interconnection and control of these subsystems.



Introduction

- **Logic design** involves determining how to **interconnect basic logic building blocks to perform a specific function**. An example of logic design is determining the interconnection of logic gates and flip-flops required to perform binary addition.
- **Circuit design** involves specifying the interconnection of specific components such as **resistors, diodes, and transistors** to form a gate, **flip-flop**, or other **logic building block**.



Binary Digits, Logic Levels, and Digital Waveforms

- Digital electronics involves **circuits** and **systems** in which there are **only two possible states**. These states are represented by two different voltage levels: A **HIGH** and a **LOW**.
- The two states can also be represented by current levels, bits and bumps on a **CD** or **DVD**, etc.
- In digital systems such as computers, combinations of the two states, called **codes**, are used to represent numbers, symbols, alphabetic characters, and other types of information.
- The two-state number system is called **binary**, and its two digits are 0 and 1. A binary digit is called a **bit**.



Binary Digits

- Each of the two digits in the binary system, **1** and **0**, is called a **bit**, which is a contraction of the words binary digit.
- In digital circuits, **two different voltage levels** are used to represent the **two bits**.
- Generally, **1** is represented by the higher voltage, which we will refer to as a **HIGH**, and a **0** is represented by the lower voltage level, which we will refer to as a **LOW**.
- The two-state number system is called **binary**, and its two digits are **0** and **1**. A binary digit is called a **bit**. This kind of system is called **positive logic**.
- Another system in which a **1** is represented by a **LOW** and a **0** is represented by a **HIGH** is called **negative logic**.



Binary Digits

- Groups of bits (combinations of **1s** and **0s**), called **codes**, are used to represent numbers, letters, symbols, instructions, and anything else required in a given application.

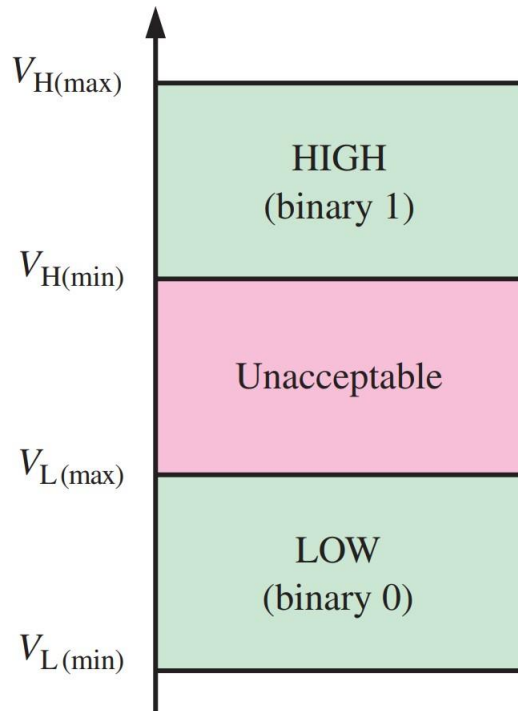


Logic Levels

- The voltages used to represent a **1** and a **0** are called **logic levels**.
- Ideally, one voltage level represents a **HIGH** and another voltage level represents a **LOW**.
- In a practical digital circuit, however, a **HIGH** can be any voltage between a specified minimum value and a specified maximum value.
- Likewise, a **LOW** can be any voltage between a specified minimum and a specified maximum.
- There can be no overlap between the accepted range of **HIGH** levels and the accepted range of **LOW** levels.



Logic Levels

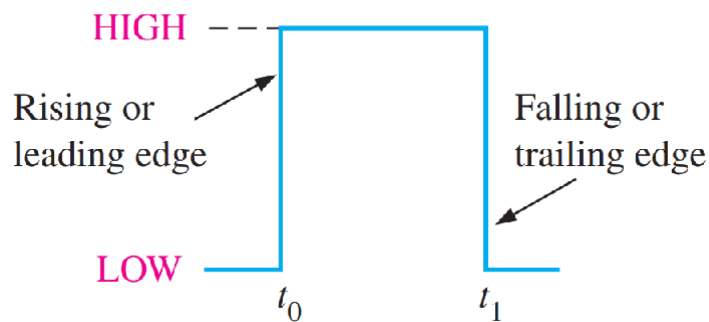


- Figure on the left side illustrates the general range of LOWs and HIGHS for a digital circuit. The variable $V_{H(max)}$ represents the maximum HIGH voltage value, and $V_{H(min)}$ represents the minimum HIGH voltage value.
- The maximum LOW voltage value is represented by $V_{L(max)}$, and the minimum LOW voltage value is represented by $V_{L(min)}$. The voltage values between $V_{L(max)}$ and $V_{H(min)}$ are unacceptable for proper operation.

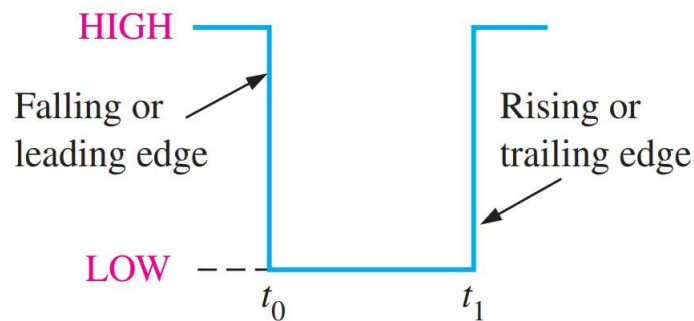


Digital Waveforms

- Digital waveforms consist of **voltage levels** that are changing back and forth between the **HIGH** and **LOW** levels or states.



- The figure on the left shows that a single **positive-going pulse** is generated when the voltage (or current) goes from its normally **LOW** level to its **HIGH** level and then back to its **LOW** level.



- The **negative-going pulse** in left Figure is generated when the voltage goes from its normally **HIGH** level to its **LOW** level and back to its **HIGH** level.

- A **digital waveform** is made up of a series of pulses.



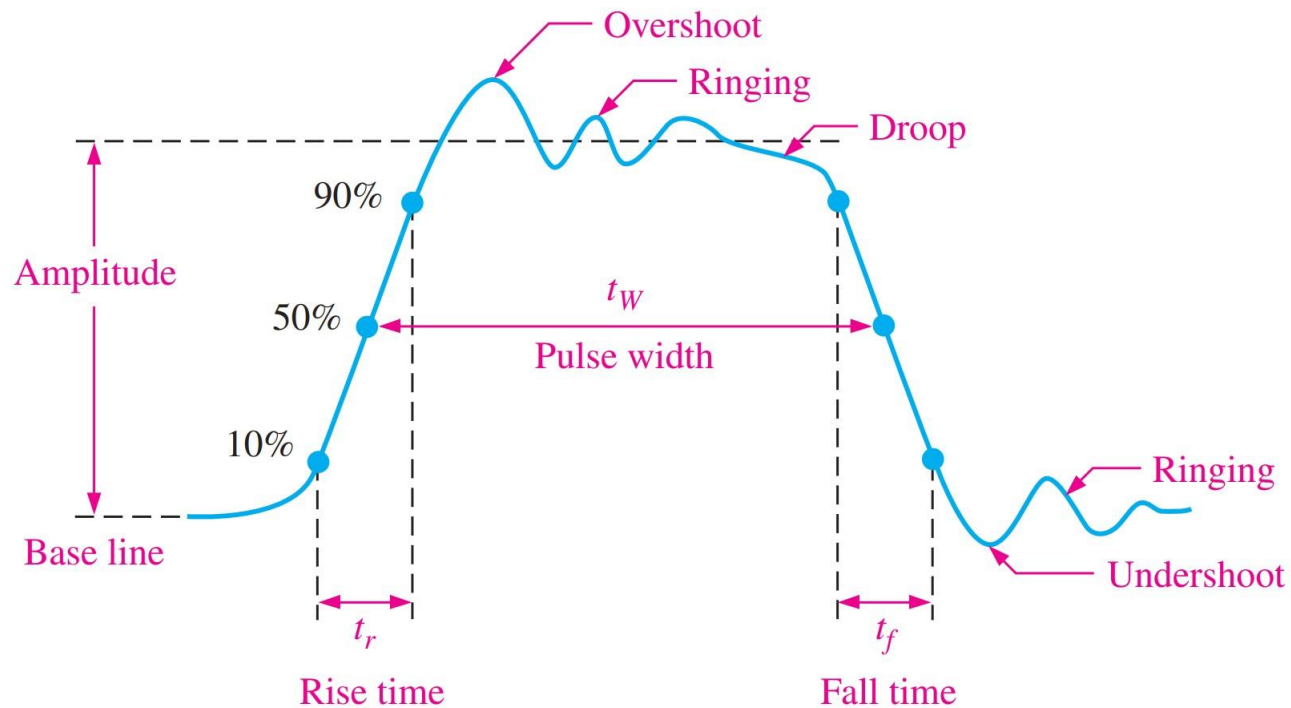
The Pulse

- A pulse has **two edges**: a **leading edge** that occurs first at time t_0 and a **trailing edge** that occurs last at time t_1 .
- For a positive-going pulse, the **leading edge is a rising edge**, and the **trailing edge is a falling edge**.
- The pulses in Figure are ideal because the rising and falling edges are assumed to change in zero time (instantaneously).
- In practice, these transitions never occur instantaneously, although for most digital work you can assume ideal pulses.



The Pulse

- Figure below shows a nonideal pulse. In reality, all pulses exhibit some or all of these characteristics. The overshoot and ringing are sometimes produced by stray inductive and capacitive effects.



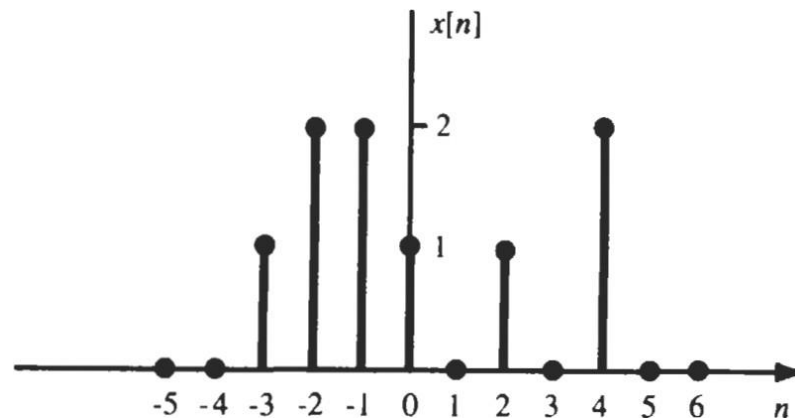
Digital Signal

- A **signal** is a function representing a **physical quantity** or **variable**, and **typically it contains information about the behavior or nature of the phenomenon**.
- For instance, in a **RC** circuit the signal may represent the voltage across the capacitor or the current flowing in the resistor. Mathematically, a signal is represented as a function of an independent variable **t** . Usually **t** represents time. Thus, a signal is denoted by **$x(t)$** .
- A **digital signal** is a signal that represents data as a sequence of **discrete values**; at any given time it can only take on, at most, one of a **finite number** of values.



Digital Signal

- If t is a discrete variable, that is, $x(t)$ is defined at discrete times, then $x(t)$ is a discrete-time signal. Since a discrete-time signal is defined at **discrete times**, a discrete-time signal is often identified as a sequence of numbers, denoted by $\{x_n\}$ or $x[n]$, where $n = \text{integer}$. Illustrations of a discrete-time signal $x[n]$ are shown below





Home Work

- Difference between Analogue and Digital
- Difference between Analogue signal and Digital signal
- Difference between Analogue Systems and Digital Systems



Next Lecture
NUMBER SYSTEMS